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Carleman Type Estimates in an Anisotropic Case and Applications

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In this paper we give proofs and some further applications of estimates announced in the author's note [5]. These results extend those of Hörmander [3] to equations with nonhomogeneous principal part such as parabolic equations and the Shrödinger equation. The basic idea was developed by Carleman in the 1930s [1] and actually all the techniques we need are available from the book [3]. Lately there were several publications dealing with nonhomogeneous principal parts. We mention, say, the papers of Dehman [2], Khalgui-Ounaies [10], Kumano-Go [12], and Saut and Scheurer [15] as well as some sections in the book of Zuily [16]. However, all these works deal with particular cases or extra regularity assumptions. We give general theorems and show how to apply them to ill-posed Cauchy problems for some important equations. Observe that hyperbolic equations were considered in the author's papers [6, 7] and in the book [8], stability was studied by F. John [9].

We would like to emphasize that the results obtained in this paper can be considered as a step in the complete understanding of the principal part of a general differential equation if we are interested in the Cauchy problem for this equation. For operators with constant coefficients such an understanding was achieved in the paper of Nirenberg [14].

Another interesting problem concerns further reduction of the smoothness assumption. Although it is impossible to do with respect to all variables, one can expect that the "time" variable requires less regularity. We mention the paper of Knabner and Vessella [11] about one-dimensional parabolic equations.

In Section 1 we formulate main results, in Section 2 we give the theory of differential quadratic forms in the anisotropic case, in Sections 3, 4 we prove Carleman estimates, and in Section 5 the stability estimate for the Cauchy problem. Section 6 is dedicated to examples.

Notation. $\mathbb{R}^n$ is the n-dimensional Euclidean space of points $x = (x_1, ..., x_n)$ with the norm $|x|$, $\zeta = (\zeta_1, ..., \zeta_n)$ is a (co-) vector in the complex

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space $\mathbb{C}^n$, and $\alpha = (\alpha_1, ..., \alpha_n)$ and $m = (m_1, ..., m_n)$ are multi-indices with natural numbers as components. We assume $0 < m_1 \leqslant \cdots \leqslant m_{q-1} < m_q = \cdots = m_n = \mu$ and we fix m;

$$|\alpha : m| = \alpha_1/m_1 + \cdots + \alpha_n/m_n, \qquad \nabla_q \phi = (0, ..., 0, \partial\phi/\partial x_q, ..., \partial\phi/\partial x_n)$$

$$D_j = -i\,\partial/\partial x_j, \qquad D^\alpha = D_1^{\alpha_1} \cdots D_n^{\alpha_n}.$$

$P(x; D) = \sum_{|\alpha : m| \leqslant 1} a_\alpha(x)\, D^\alpha$ is a linear partial differential operator with (complex-valued) coefficients a_α,

$$\zeta^\alpha = \zeta_1^{\alpha_1} \cdots \zeta_n^{\alpha_n}, \qquad |\zeta|_m = (|\zeta_1|^{2m_1} + \cdots + |\zeta_n|^{2m_n})^{1/2}.$$

$P(x; \zeta) = \sum_{|\alpha : m| \leqslant 1} a_\alpha(x)\, \zeta^\alpha$ is the symbol of P and the same sum over $|\alpha : m| = 1$ is its m-principal part $P_m(x; \zeta)$;

$$P^{(j)}(x, \zeta) = \partial P(x; \zeta)/\partial \zeta_j, \qquad P_j(x; \zeta) = \partial P(x; \zeta)/\partial x_j.$$

Also we introduce special anisotropic norms

$$\|u\|_{m;\theta}(\Omega) = \left(\sum_{|\alpha : m| \leqslant \theta} \int_\Omega |D^\alpha u|^2\, dx \right)^{1/2}$$

$$|||u|||_\theta = \left(\int |\hat{u}(\xi)|^2\, |\xi + i\tau N|_m^{2\theta}\, d\xi \right)^{1/2},$$

where $\hat{u}(\xi)$ denotes the Fourier transform of u and N is to be defined in Sections 3 and 4.

By C we denote positive constants depending only on P, Ω, and ϕ and which are different in different places.

1. Main Results

Let Ω be a bounded open subset of $\mathbb{R}^n$. Assume that

$$\text{a (real-valued) function } \phi \in C^2(\operatorname{cl}\Omega), \qquad \nabla_q \phi \neq 0 \text{ on } \operatorname{cl}\Omega. \tag{1.1}$$

The coefficients a_α of the operator P are supposed to satisfy the regularity conditions

$$a_\alpha \in C^1(\operatorname{cl}\Omega) \quad \text{if} \quad |\alpha : m| = 1, \text{ otherwise } a_\alpha \in L_\infty(\Omega). \tag{1.2}$$

Theorem 1.1. *Suppose that*

$$P_m(x; \xi) \neq 0 \qquad \textit{for all} \quad \operatorname{cl}\Omega \textit{ and } \xi \in \mathbb{R}^n \backslash \{0\}. \tag{1.3}$$

Suppose that the conditions

$$P_m(x;\zeta)=0, \qquad x\in \operatorname{cl}\Omega,\ \zeta=\xi+i\tau\nabla_q\phi\neq 0 \tag{1.4}$$

imply that

$$\begin{gathered}\sum \partial^2\phi/\partial x_j\,\partial x_k P_m^{(j)}(x;\zeta)\,\overline{P_m^{(k)}(x;\zeta)}\\ +\tau^{-1}\operatorname{Im}\sum P_{mk}(x;\zeta)\,\overline{P_m^{(k)}(x;\zeta)}>0\\ (\textit{the sums are over } j, k=q, ..., n).\end{gathered} \tag{1.5}$$

Then there is a constant C such that

$$\begin{aligned}&\tau^{2\mu(1-|\alpha:m|)}\int_\Omega |D^\alpha u|^2\exp(2\tau\phi)\,dx\\ &\qquad\leqslant C\tau\int_\Omega |Pu|^2\exp(2\tau\phi)\,dx \qquad \textit{if}\quad |\alpha:m|\leqslant 1\end{aligned} \tag{1.6}$$

for all $u\in C_0^\infty(\Omega)$ *and* $\tau>C$.

THEOREM 1.2. *Suppose that*

$$\textit{the coefficients } a_\alpha \textit{ with } |\alpha:m|=1 \textit{ are real-valued.} \tag{1.7}$$

Suppose that the condition "(1.4) *implies* (1.5)" *is satisfied and that moreover the equalities*

$$P_m(x;\xi)=0, \qquad \sum_{q\leqslant j} P_m^{(j)}(x;\xi)\,\partial\phi/\partial x_j=0,\ \xi\in\mathbb{R}^n\backslash\{0\} \tag{1.8}$$

imply that

$$\begin{gathered}\sum \partial^2\phi/\partial x_j\,\partial x_k P_m^{(j)}(x;\xi)\,P_m^{(k)}(x;\xi)\\ +\sum \left(P_{mj}^{(k)}(x;\xi)\,P_m^{(j)}(x;\xi)-P_{mj}(x;\xi)\,P_m^{(jk)}(x;\xi)\right)\partial\phi/\partial x_j>0\\ (\textit{the sums are over } j, k=q, ..., n)\end{gathered} \tag{1.9}$$

for $x\in\operatorname{cl}\Omega$.

Then there is a constant C such that the estimate (1.6) *is valid for all* α *with* $|\alpha:m|<1$.

We give applications to the uniqueness and stability of the Cauchy problem.

Define $\Omega_\varepsilon = \Omega \cap \{\varepsilon < \phi\}$ and suppose that $\Gamma_0 = \partial\Omega_0 \setminus \{\phi = 0\}$ is a hypersurface in $\mathbb{R}^n$ of class C^μ. Consider the Cauchy problem

$$\begin{aligned} Pu &= f \qquad \text{on } \Omega_0 \\ (\partial/\partial\nu)^j u &= g_j \qquad \text{on } \Gamma_0,\ j=0, ..., \mu-1 \\ \|u\|_{m;\,1}(\Omega_0) &< +\infty, \end{aligned} \tag{1.10}$$

where ν is the unit normal to Γ_0.

THEOREM 1.3. *If the function u is a solution to the Cauchy problem* (1.10) *then under the conditions of Theorem* 1.1 *we have*

$$\|u\|_{m;\,1}(\Omega_\varepsilon) \leqslant C(\varepsilon)(\delta^{\kappa(\varepsilon)}\, \|u\|_{m;\,0}^{1-\kappa(\varepsilon)}\,(\Omega_0) + \delta) \tag{1.11}$$

and under the conditions of Theorem 1.2 *we have*

$$\|u\|_{m;\,1-1/\mu}(\Omega_\varepsilon) \leqslant C(\varepsilon)(\delta^{\kappa(\varepsilon)}\, \|u\|_{m;\,1-1/\mu}^{1-\kappa(\varepsilon)}\,(\Omega_0) + \delta), \tag{1.12}$$

where $\varepsilon > 0$, $\kappa(\varepsilon) = \varepsilon/(2\Phi - \varepsilon)$, $\Phi = \sup \phi$ over Ω_0, and

$$\delta = \|f\|_{(0)}\,(\Omega_0) + \|g_0\|_{\mu-1/2}\,(\Gamma_0) + \cdots + \|g_{\mu-1}\|_{1/2}\,(\Gamma_0).$$

Theorem 1.3 says that a solution u to the Cauchy problem (1.10) is unique and moreover that it is stable on any Ω_ε provided it is bounded on Ω_0.

Now we will show that the conditions of Theorems 1.1, 1.2 are invariant under the changes of coordinates $x = x(x^*)$ such that $x_1 = x_1^*, ..., x_{q-1} = x_{q-1}^*$. This is essential for proofs.

Denote by H_q the left side of the inequality (1.6) and by H the same expression with the sums over $j, k = 1, ..., n$. If w is a function then $P_m(x; \nabla w(x))$ is invariant under our specific changes of coordinates, so repeating the argument from Hörmander's book [3, p. 186], we conclude that H is invariant as well.

Let P^*, H^* be the operator P in $*$-coordinates and the related form H. Let $P_m^*(x^*; \zeta^*) = 0$, $\zeta^* = \xi^* + i\tau \nabla\phi^*$, then due to the invariancy $P_m(x; \zeta) = 0$ and from the condition "(1.4) implies (1.5)" we have $H_q(x; \zeta) > 0$. Let

$$\zeta(t) = (t^{1/m_1}\zeta_1, ..., t^{1/m_n}\zeta_n)$$

then $P_m(x; \zeta(t))$, $H_q(x; \zeta(t))$ are linear in t. Since $H(x; \zeta(t)) - H_q(x; \zeta(t))$ is bounded by $Ct^{1-1/\mu}$, $t > 0$, we have $H(x; \zeta(t)) > 0$ for large t and due to the invariancy the same inequality is valid for

$$H^*(x^*; \zeta^*(t)) = H_q^*(x^*; \zeta^*)t + O(t^{1-1/\mu})$$

hence $H_q^*(x^*; \zeta^*) > 0$. So we proved that the condition of Theorem 1.1 is $x - x^*$ invariant.

Observing that the condition "(1.8) implies (1.9)" is the limit of the condition "(1.4) implies (1.5)" as τ goes to zero we obtain the invariancy of this condition as well (compare with [3, pp. 193–194]).

2. Differential Quadratic Forms

This section is purely technical. It transfers the arguments of Hörmander [3, Sect. 8.2] onto the anisotropic case we deal with.

Introduce the differential quadratic form

$$F(x, D, \bar{D})\, u\bar{u} = \sum a_{\alpha\beta}(x)\, D^\alpha u\, \overline{D^\beta u} \tag{2.1}$$

with the coefficients $a_{\alpha\beta}$ and introduce its symbol

$$F(x, \zeta, \bar{\zeta}) = \sum a_{\alpha\beta}(x)\, \zeta^\alpha \bar{\zeta}^\beta. \tag{2.2}$$

We assume that these sums are finite.

We call the pair (λ, σ) the m-order of F if $|(\alpha+\beta) : m| \leqslant \lambda$, $|\alpha : m| \leqslant \sigma$, $|\beta : m| \leqslant \sigma$ for all (non-zero) coefficients $a_{\alpha\beta}$ of F.

Lemma 2.1. *Let $F(D, \bar{D})\, u\bar{u}$ be a differential quadratic form of m-order (λ, σ) with constant coefficients. Suppose that*

$$F(\xi, \xi) = 0 \qquad \text{if} \quad \xi \in \mathbb{R}^n. \tag{2.3}$$

Then there are differential quadratic forms $G^k(D, \bar{D})\, u\bar{u}$ such that

$$\int F(D, \bar{D})\, u\bar{u} = \int \sum \partial/\partial x_k G^k(D, \bar{D})\, u\bar{u}, \tag{2.4}$$

the sum over $k = 1, ..., n$. Moreover,

$$G^k(\xi, \xi) = -1/2\, \partial/\partial\eta_k F(\xi + i\eta, \xi - i\eta)\,|_{\eta=0}, \qquad \xi \in \mathbb{R}^n \tag{2.5}$$

and it is possible to choose G of m-order $(\lambda - 1/\mu, \sigma)$ and if $\lambda < 2\sigma$ then of m-order $(\lambda - 1/\mu, \sigma - 1/\mu)$.

Proof. Existence of G^k was proved in [3, Lemma 8.2.1], where it was observed that the equality (2.4) is equivalent to the relation

$$F(\zeta, \bar{\zeta}) = i \sum (\zeta_k - \bar{\zeta}_k)\, G^k(\zeta, \bar{\zeta}) \qquad \text{(the sum over } k = 1, ..., n). \tag{2.6}$$

We have only to check the claim about the m-order.

We write $F_1 \equiv F_2$ if the difference $F = F_1 - F_2$ has the form (2.6) with G^k of m-order $(\lambda - 1/\mu, \sigma - 1/\mu)$ if $\lambda < 2\sigma$ and of m-order $(\lambda - 1/\mu, \sigma)$ if $\lambda = 2\sigma$. We claim that

$$\zeta^\alpha \bar{\zeta}^\beta \equiv \zeta^{\alpha(*)} \bar{\zeta}^{\beta(*)} \tag{2.7}$$

if $\alpha + \beta = \alpha(*) + \beta(*)$ and if both sides are of m-order (λ, σ).

Consider the case when $\lambda < 2\sigma$. Then either $|\alpha : m| < \sigma$ or $|\beta : m| < \sigma$ and using the identities

$$\bar{\zeta}_j = \zeta_j - (\zeta_j - \bar{\zeta}_j) \qquad \text{or} \qquad \zeta_j = \bar{\zeta}_j + (\zeta_j - \bar{\zeta}_j)$$

one can show that replacing $\bar{\zeta}_j$ by ζ_j or ζ_j by $\bar{\zeta}_j$ in the left side of (2.7) does not change the $\equiv$ relation. By repeating this procedure we obtain (2.7).

Consider the case when $\lambda = 2\sigma$. Then the above argument is valid unless

$$|\alpha : m| = |\beta : m| = |\alpha(*) : m| = |\beta(*) : m| = \sigma,$$

if we use the identity

$$\zeta_j \zeta_k - \bar{\zeta}_k \bar{\zeta}_j = (\zeta_j - \bar{\zeta}_j)\,\zeta_k - (\zeta_k - \bar{\zeta}_k)\,\bar{\zeta}_j$$

and replace simultaneously two factors in the left side of (2.7).

Summing up we obtain the relation (2.7).

From this relation it follows that $F \equiv F_0$ where F_0 is of the form (2.1) with no two different terms with the same sum $\alpha + \beta$. Due to the definition of $\equiv$ we have $F_0(\xi, \xi) = F(\xi, \xi) = 0$ by the condition (2.3), so $F_0 \equiv 0$.

The proof is complete.

LEMMA 2.2. *Let $F(x, D, \bar{D})\, u\bar{u}$ be a differential quadratic form of m-order (λ, σ) with $C^1(\bar{\Omega})$-coeffcients. Suppose that the relation* (2.3) *is valid for any $x \in \Omega$.*

Then there is a differential quadratic form $G(x, D, \bar{D})\, u\bar{u}$ with coefficients in $C(\bar{\Omega})$ such that

$$\int F(x, D, \bar{D})\, u\bar{u}\, dx = \int G(x, D, \bar{D})\, u\bar{u}\, dx, \qquad u \in C_0^\infty(\Omega). \tag{2.8}$$

Moreover, G is of m-order $(\lambda - 1/\mu, \sigma)$ if $\lambda < 2\sigma$ is of m-order $(\lambda - 1/\mu, \sigma - 1/\mu)$ and

$$G(x, \xi, \xi) = 1/2 \sum \partial^2/\partial x_k\, \partial \eta_k F(x, \xi + i\eta, \xi - i\eta)\,|_{\eta = 0}$$

$$\text{(the sum over } k = 1, \ldots, n). \tag{2.9}$$

The proof is similar to the one of Lemma 8.2.2 in [3] if we use Lemma 2.1 instead of Lemma 8.2.1 in [3].

3. Proof of Theorem 1.1

We give two simple auxiliary results.

LEMMA 3.1. *If for any $a \in \Omega$ and some $\varepsilon > 0$ the estimate* (1.6) *is valid for all $u \in C_0^\infty(\Omega \cap B(a; \varepsilon))$ then it is valid for all $u \in C_0^\infty(\Omega)$.*

LEMMA 3.2. *If the estimate* (1.6) *is valid for $P = P_m$ then it is valid for any P.*

To prove Lemma 3.1 it suffices to use the partition of unity and to choose τ large.

Lemma 3.2 follows directly from the structure of the estimate (1.6) for large τ. In the isotropic case $(m = (m_1, ..., m_n))$ proofs are given in Section 8.3 of Hörmander's book [3].

Proof of Theorem 1.1. In view of Lemmas 3.1 and 3.2 we may assume that $\Omega = B(0; \varepsilon)$ where ε can be chosen arbitrarily small and positive. By the condition (1.1) we have $\nabla_q \phi(0) \neq 0$, so changing (and relabeling) coordinates as at the end of Section 1 we may assume that $\phi(x) = x_n$.

Letting

$$v(x) = u(x) \exp(-\tau x_n)$$

we obtain

$$\begin{aligned} &\int |P_m(x; D) u|^2 \exp(2\tau x_n)\, dx \\ &\quad = \int |P_m(x; D + irN) v|^2\, dx \\ &\quad \geqslant \int (|P_m(x; D + i\tau N) v|^2 - |\bar{P}_m(x; D - i\tau N) v|^2)\, dx \\ &\quad = \int F(x, D, \bar{D}, \tau)\, v\bar{v}\, dx, \end{aligned} \tag{3.1}$$

where $N = (0, ..., 0, 1)$ and

$$\begin{aligned} F(x; \zeta, \bar{\zeta}, \tau) &= |P_m(x; \zeta + i\tau N)|^2 - |P_m(x; \bar{\zeta} + i\tau N)|^2 \\ &= \sum \tau^j F_j(x; \zeta, \bar{\zeta}) \end{aligned}$$

(the sum over $j=0, \ldots, 2\mu-1$) for certain forms F_j of m-order $(2-j/\mu; 1)$ with $F_j(x; \xi, \xi)=0$. From Lemma 2.2 we derive that there are differential quadratic forms G_j of m-order $(2-(j+1)/\mu, 1)$ and with coefficients from $C(\bar{\Omega})$ such that

$$\int F_j v\bar{v} = \int G_j v\bar{v}.$$

Letting

$$G = \sum \tau^j G_j \tag{3.2}$$

we get

$$\int F v\bar{v} = \int G v\bar{v} \tag{3.3}$$

and

$$G(x; \xi, \xi, \tau) = 2\,\mathrm{Im}\Big(\sum \big(P_{m,k}(x; \xi + i\tau N)\,\overline{P_m^{(k)}(x; \xi + i\tau N)}$$

$$+ P_m(x; \xi + i\tau N)\,\overline{P_{m,k}^{(k)}(x; \xi + i\tau N)}\big)\Big)$$

$$\text{(the sum over } k = 1, \ldots, n). \tag{3.4}$$

We claim that for certain positive constants C_1, C_2 we have

$$|\xi + i\tau N|_m^2 \leqslant C_1 \tau G(0, \xi, \xi, \tau) + C_2\, |P_m(0, \xi + i\tau N)|^2 \tag{3.5}$$

for $\tau \geqslant C_1$, $\xi \in \mathbb{R}^n$.

To prove (3.5) denote by $G^*(x, \zeta, \bar{\zeta}, \tau)$ the symbol that is the sum of terms $a_{\alpha\beta}(x)\,\zeta^\alpha \bar{\zeta}^\beta \tau^j$ of $G(x, \zeta, \bar{\zeta}, \tau)$ with $|(\alpha+\beta) : m| + j/\mu = 2 - 1/\mu$. All terms of $G - G^*$ involve differentiations of $P_m(x; \zeta)$ with respect to $\zeta_1, \ldots, \zeta_{q-1}$ when we lose more of the anisotropic power as when differentiating with respect to $\zeta_q, \ldots, \zeta_n$, so the symbol $(G-G^*)(0, \xi, \xi, \tau)$ is the sum of terms

$$a(x)\, \xi_1^{\gamma_1} \cdots \xi_n^{\gamma_n} \tau^j$$

with $|\gamma : m| + j/\mu \leqslant 2 - 2/\mu$. Hence

$$|G - G^*|\,(0, \xi, \xi, \tau) \leqslant C\, |\xi + i\tau N|_m^{2-2/\mu}. \tag{3.6}$$

Show that there are constants C_3, C_4 such that

$$|\xi + i\tau N|_m^2 \leqslant C_3 \tau G^*(0, \xi, \xi, \tau) + C_4\, |P_m(0; \xi + i\tau N)|^2 \tag{3.7}$$

for $\tau \geqslant 0$, $\xi \in \mathbb{R}^n$. Both parts of this inequality are m-homogeneous so we may assume $|\xi + i\tau N|_m = 1$. Since by the condition $P_m(0; \xi) \neq 0$ if $\xi \in \mathbb{R}^n \setminus \{0\}$ we have $\tau \geqslant \tau_0 > 0$ on the compact set K_0 where

$$K_\delta = \{(\xi, \tau) : |P_m(0, \xi + i\tau N)| \leqslant \delta,\ |\xi + i\tau N|_m = 1\}.$$

Due to the definition of G^* its symbol at $\zeta = \xi$, $\zeta = \xi$ is given by the expression (3.4) with the sum over $k = q, ..., n$, so the pseudo-convexity condition (1.5) gives $G^*(0, \xi, \xi, \tau) \geqslant 1/C$ on K_0. By the continuity argument we conclude that $G^* \geqslant 1/(2C)$ on K_δ for certain $\delta > 0$. If $(\xi, \tau) \notin K_\delta$ then we obtain (3.7) by choosing C_4 large. If $(\xi, \tau) \in K_\delta$ then (3.7) is valid with $C_3 = C/(2\tau_0)$.

The estimate (3.5) follows from the estimates (3.6) and (3.7) if τ is large enough.

Multiplying both parts of (3.7) by $|\hat{v}(\xi)|^2$ and integrating in ξ we obtain

$$|||v|||_1^2 \leqslant C_1 \tau \int G(0, D, \bar{D}, \tau)\, v\bar{v}\, dx + C_2 \int |P_m(0, D + i\tau N)v|^2\, dx. \quad (3.8)$$

From the definition it follows that $\tau(G - G^*)(x, \zeta, \bar{\zeta}, \tau)$ is the sum of terms $a(x)\,\zeta^\alpha \bar{\zeta}^\beta \tau^{j+1}$ with $|(\alpha + \beta) : m| \leqslant 2 - (j+1)/\mu$, $|\alpha : \mu| \leqslant 1$, $|\beta : \mu| \leqslant 1$, and with continuous coefficients $a(x)$, $a(0) = 0$, so

$$\tau \int (G(x, D, \bar{D}, \tau) - G^*(x, D, \bar{D}, \tau)\, v\bar{v})\, dx \leqslant \omega(\varepsilon)\, |||v|||_1^2. \quad (3.9)$$

A similar estimate is valid for $P(x, D + i\tau N) - P(0, D + i\tau N)$. Thus from (3.8) we have

$$|||v|||_1^2 \leqslant C_1 \tau \int G(x, D, \bar{D}, \tau)\, v\bar{v}\, dx$$
$$+ C_2 \int |P(x, D + i\tau N)v|^2\, dx + 2\omega(\varepsilon)\, |||v|||_1^2. \quad (3.10)$$

Choosing ε so small that $2\omega(\varepsilon) < 1/2$ we remove the last term. Using the inequality (3.1) and the relation (3.3) we remove the term with G. From Minkovsky's inequality we have

$$\tau^{2\mu(1 - |\alpha : m|)} \xi_1^{2\alpha_1} \cdots \xi_n^{2\alpha_n} \leqslant C(\xi_1^{2m_1} + \cdots + \xi_{n-1}^{2m_{n-1}} + (\xi_n^{2m_n} + \tau^{2m_n}))$$

so the definition of $|||\ |||$-norm and Parseval's equality give

$$\tau^{2\mu(1 - |\alpha : m|)} \int |D^\alpha v|^2 \leqslant C\, |||v|||_1^2$$

if $|\alpha : m| \leqslant 1$. Now the estimate (1.6) follows from (3.10) with some terms removed as above.

The proof is complete.

4. Proof of Theorem 1.2

As in the proof of Theorem 1.1 we may assume that $\Omega = B(0; \varepsilon)$ and $\phi(x) = x_n$.

Letting

$$v(x) = u(x)\exp(\tau x_n) \tag{4.1}$$

as above, with a constant A to be chosen later, we obtain

$$\begin{aligned}&\int |P_m(x; D)u|^2 \exp((\tau + A)\, x_n)\, dx\\ &\quad = \int |P_m(x; D + i\tau N)v|^2 \exp(2Ax_n)\, dx \geqslant \int F(x, D, \bar{D}, \tau)\, v\bar{v}\, dx,\end{aligned} \tag{4.2}$$

where the form F is defined by its symbol

$$\begin{aligned}F(x, D, \bar{D}, A, \tau) &= \exp(2Ax_n)(|P_m(x, \zeta + i\tau N)|^2 - |P_m(x, \zeta - i\tau N)|^2)\\ &= \sum \tau^j F_j(x, \zeta, \bar{\zeta}, A)\end{aligned} \tag{4.3}$$

for certain form F_j of m-order $(2 - j/\mu; 1)$, $j = 1, ..., 2\mu - 1$. Apparently, $F_j(x, \xi, \xi, A) = 0$, so according to Lemma 2.2 there are forms $G_j(x, D, \bar{D}, A)\, v\bar{v}$ of m-order $(2 - (j+1)/\mu,\ 1 - 1/\mu)$ with continuous coefficients such that

$$\int G_j(x, D, \bar{D}, A)\, v\bar{v}\, dx = \int F_j(x, D, \bar{D}, A)\, v\bar{v}\, dx. \tag{4.4}$$

Letting $G = G_1 + \cdots + \tau^{2\mu - 2} G_{2\mu - 1}$ and $P = P_m$ as in the book of Hörmander [3, p. 196], we obtain

$$\begin{aligned}G(0, \xi, \xi, A, \tau) = 2\tau^{-1}\,\mathrm{Im} \sum \Big(&P_j(0, p + i\tau N)\, \overline{P^{(j)}(0, \xi + i\tau N)}\\ &+ P(0, \xi + i\tau N) \sum \overline{P_j^{(j)}(0, \xi + i\tau N)}\\ &+ 2AP(0, \xi + i\tau N) \sum \overline{P^{(j)}(0, \xi + i\tau N)}\Big)\end{aligned} \tag{4.5}$$

and, if τ goes to 0,

$$G(0, \xi, \xi, A, 0) = 2\left(\sum (P_j^{(n)}(0, \xi)\, P^{(j)}(0, \xi) - P_j(0, \xi)\, P^{(jn)}(0, \xi))\right.$$
$$\left. + P^{(n)}(0, \xi)\left(\sum P_j^{(j)}(0, \xi) + 2AP^{(n)}(0, \xi)\right)\right) \qquad (4.6)$$

if $\xi \in \mathbb{R}^n$ and $P(0, \xi) = 0$. Here the sums are over $j = 1, ..., n$.

We claim that for certain positive constants A, C_1, C_2 we have

$$|\xi + i\tau N|_m^{2-2/\mu} \leqslant C_1 G(0, \xi, \xi, A, \tau) + C_2\, |P(0, \xi + i\tau N)|^2\, |\xi + i\tau N|_m^{-2/\mu} \qquad (4.7)$$

if $\tau \geqslant C_1$, $\xi \in \mathbb{R}^n$.

Denote by $G^*(x, \zeta, \bar{\zeta}, A, \tau)$ the symbol formed by terms $a(x)\, \zeta^\alpha \bar{\zeta}^\beta \tau^j$ of the symbol $G(x, \zeta, \zeta, A, \tau)$ with $|(\alpha + \beta) : m| + j/\mu = 2 - 2/\mu$, then as in the proof of Theorem 1.1 we have the estimate

$$|G - G^*|\ (0, \xi, \xi, A, \tau) \leqslant C\, |\xi + i\tau N|_m^{2-3/\mu} \qquad (4.8)$$

and the formulae (4.5), (4.6) for G^* are valid provided the sums are taken over $j = q, ..., n$ instead of $1, ..., n$.

We will show that there are constants C_3, C_4 such that

$$|\xi + i\tau N|_m^{2-2/\mu} \leqslant C_3 G^*(0, \xi, \xi, A, \tau) + C_4\, |P(0, \xi + i\tau N)|^2\, |\xi + i\tau N|_m^{-2/\mu}. \qquad (4.9)$$

Due to m-homogeneity we may assume that $|\xi + i\tau N|_m = 1$. Introduce the compact set

$$K(\delta, \varepsilon) = \{\xi : |\xi|_m = 1,\ |P(0, \xi)| \leqslant \delta,\ |P^{(n)}(0, \xi)| \leqslant \varepsilon\}.$$

From the pseudo-convexity condition "(1.7) implies (1.8)" and from the formula (4.6) for G^* it follows that $3/C_3 \leqslant G^*$ for certain C_3 so $2/C_3 \leqslant G^*$ on $K(0, \varepsilon)$ for some ε. If $|P^{(n)}(0, \xi)| \geqslant \varepsilon$ then we can always find A to satisfy $2/C_3 \leqslant G^*$ as well, so we obtain this inequality on the compact set $K(\delta, +\infty)$ for some δ. If $|\xi|_m = 1$, $|P(0, \xi)| \geqslant \delta$ then we obtain (4.9) by choosing C_4 large. Due to continuity and compactness reasons we conclude that (4.9) remains to be valid, probably, with larger C_3, C_4 if $|\tau| \leqslant \tau_0$ where τ_0 is positive and small. On the set $\{(\xi, \tau) : |\xi + i\tau N|_m = 1,\ |\tau| \geqslant \tau_0\}$ the proof of (4.9) is similar to the one in Theorem 1.1.

Also as above, from (4.8), (4.9) we obtain (4.7).

Multiplying both sides of (4.7) by $|\hat{v}(\xi)|^2$, integrating, and using the definition of the norm $|||\ |||$ we get

$$|||v|||^2_{1-1/\mu} \leqslant C_1 \int G(0, D, \bar{D}, A, \tau)\, v\bar{v}\, dx$$
$$+ C_2\, |||P(0, D+i\tau N)v|||^2_{-1/\mu}. \tag{4.10}$$

Since G_j is of m-order $(2-(j+1)/\mu,\ 1-1/\mu)$ we conclude that

$$\int G(0, D, \bar{D}, A, \tau)\, v\bar{v}\, dx$$
$$\leqslant \int G(x, D, \bar{D}, A, \tau)\, v\bar{v}\, dx + \omega(\varepsilon)\, |||v|||^2_{(1-1/\mu)}, \tag{4.11}$$

where $\omega(\varepsilon)$ goes to 0 as ε goes to 0.

To estimate the difference between $P(0;\ )$ and $P(x;\ )$ we need the following lemma.

LEMMA 4.1. *Let* $b \in \mathrm{Lip}(B(0; \varepsilon))$, $b(0)=0$. *Then there is a function* $\omega(\varepsilon, \tau)$ *that goes to* 0 *as* $\varepsilon + 1/r$ *goes to* 0 *such that*

$$|||b(D+i\tau N)^\alpha u|||_{-1/\mu} \leqslant \omega(\varepsilon, \tau)\, |||u|||_{1-1/\mu}$$

for all functions $u \in C_0^\infty(B(0; \varepsilon))$ *and for all multi-indices* α *with* $|\alpha : m| \leqslant 1$.

Proof. We may assume $\alpha_j > 0$ for some j. Let $D^* = D + i\tau N$, $(D^*)^\alpha = D_j^*(D^*)^\beta$, and let $v = (D^*)^\beta u$. From the definition of the norm $|||\ |||$ it follows that

$$|||v|||_{1/m_j - 1/\mu} \leqslant C\, |||u|||_{1-1/\mu}$$

so it suffices to show that

$$|||bD_j^* v|||_{-1/\mu} \leqslant \omega(\varepsilon, \delta)\, |||v|||_{1/m_j - 1/\mu}.$$

We will prove the more general inequality

$$|||bD_j^* v|||_{-\theta} \leqslant \omega(\varepsilon, \tau)\, |||v|||_{1/m_j - \theta} \tag{4.12}$$

for $0 \leqslant \theta \leqslant 1/m_j$.

We consider two extremal cases $\theta = 0$ and $1/m_j$, and then apply interpolation.

For $\theta = 0$ we have

$$||bD_j^* v||_0 \leqslant C\varepsilon\, |||D_j^* v|||_0 \leqslant \varepsilon\, |||v|||_{1/m_j}$$

due to the assumptions $b(0)=0$, $\operatorname{supp} v \subset B(0;\varepsilon)$ and to the definition of the norm $|||\ |||$.

For $\theta=1/m_j$ we observe that $bD_j^*v = D_j(bv)-(D_jb)v$, s

$$|||bD_j^*v|||_{-\theta} \leqslant |||D_j^*(bv)|||_{-\theta} + |||(D_jb)v|||_{-\theta}.$$

From the definition of the norm $|||\ |||$ and from the inequality $|\zeta|_m \leqslant \tau^\mu$ we have

$$|||(D_jb)v|||_{-\theta} \leqslant Cr^{-\theta\mu}\ |||v|||_0, \qquad C=\sup |D_jb|$$

thus it suffices to bound $D_j^*(bv)$. We have

$$|||D_j^*(bv)|||_{-1/m_j} = \int |\zeta|_m^{-2/m_j}\ |\xi_j + i\tau N_j|^2\ |bv|^2\ d\xi$$

$$\leqslant \int |bv|^2\ d\xi = |||bv|||_0 \leqslant C\varepsilon\ |||v|||_0.$$

Summing up we obtain the estimate (4.12) with $\omega(\varepsilon,\tau)=C(\varepsilon+\tau^{-\mu/m_j})$ for $\theta=0$ and $1/m_j$. Applying standard interpolation methods (Lions and Magenes [13, Theorem 5.1]) we obtain (4.12) also for all intermediate values of θ.

The proof is complete.

Now we complete the proof of Theorem 1.2.

From Lemma 4.1 it follows that

$$|||(P(0;D+i\tau N)-P(x;D+i\tau N))v|||_{-1/\mu}^2 \leqslant \omega(\varepsilon,\tau)\ |||v|||_{1-1/\mu}^2.$$

Using (4.10) and (4.11) we obtain

$$|||v|||_{(1-1/\mu)}^2 \leqslant C_1 \int G(x,D,\bar{D},A,\tau)\, v\bar{v}\, dx$$

$$+ C_2\ |||P(x;D+i\tau N)v|||_{-1/\mu} + \omega(\varepsilon,\tau)\ |||v|||_{1-1/\mu}^2.$$

Choosing ε and τ so that $\omega(\varepsilon,\tau)<1/2$ we eliminate the last term. Observing that, due to (4.2) and (4.4),

$$\tau \int G(x,D,\bar{D},\tau,A)\, v\bar{v}\, dx \leqslant \int |P(x;D+i\tau N)v|^2 \exp(2Ax_n)\, dx$$

so that

$$|||P(x;D+i\tau N)v|||_{-1/\mu} \leqslant C\tau^{-1}\ |||P(x;D+i\tau N)v|||_0$$

we complete the proof of the estimate (1.6) for $|\alpha : m| < 1$ as above in Section 3.

5. Proof of Theorem 1.3

From Extension Theorems for Sobolev spaces the existence of a function w follows such that

$$(\partial/\partial N)^j w = g_j,\ j \leqslant \mu - 1, \qquad \|D^\alpha w\|_{m;0}(\Omega_0) \leqslant C\delta,\ |\alpha| \leqslant s. \tag{5.1}$$

Let

$$u_0 = u - w, \tag{5.2}$$

then

$$Pu_0 = f - Pw \text{ in } \Omega_0, \qquad (\partial/\partial N)^j u_0 = 0 \text{ on } \Gamma_0,\ |\alpha| \leqslant \mu - 1. \tag{5.3}$$

Let $\psi \in C_0^\infty(\mathbb{R}^n)$, $0 \leqslant \psi \leqslant 1$, $\psi = 1$ on $\Omega_{\varepsilon/2}$, $\psi = 0$ on $\Omega_0 \backslash \Omega_{\varepsilon/4}$. Due to (5.3) the function u_0 has zero Cauchy data on Γ_0, so the function $v = \psi u_0$ has zero Cauchy data on $\partial\Omega_0$ and therefore it can be approximated by $C_0^\infty(\Omega_0)$-functions in the norm $\| \ \|_{m;1}$.

Using Leibniz's formula we get

$$|P(\psi u_0)|^2 \leqslant 2\,|Pu_0|^2 + C(\varepsilon) \sum_{|\alpha:m|<1} |D^\alpha u_0|^2 \qquad \text{on } \Omega_0. \tag{5.4}$$

Under the conditions of Theorem 1.1 the estimate (1.6) is valid for all $C_0^\infty(\Omega_0)$-functions and therefore for their limits described above. Using (5.4) we obtain

$$\sum_{|\alpha:m|<1} \tau^{2\mu(1-|\alpha:m|)} \int_{\Omega_0} |D^\alpha(\psi u_0)|^2 \exp(2\tau\phi)\,dx$$
$$\leqslant C\tau \int_{\Omega_0} |Pu_0|^2 \exp(2\tau\phi)\,dx + C(\varepsilon)\tau \sum_{|\alpha:m|<1} \int_{\Omega_0} |D^\alpha u_0|^2 \exp(2\tau\phi)\,dx.$$

Shrinking the integration domain in the left side to $\Omega_{\varepsilon/2}$, using that $\psi = 1$ on this set, splitting the last integral into integrals over $\Omega_{\varepsilon/2}$ and over $\Omega_0 \backslash \Omega_{\varepsilon/2}$, and choosing $\tau > 2C(\varepsilon)$ we bound the sum of the integrals from the right side by 1/2 of the sum from the left side, then shrinking the integration domain in the left side once more to Ω_ε we get

$$\sum_{|\alpha:m|<1} \tau^{2\mu(1-|\alpha:m|)} \int_{\Omega_\varepsilon} |D^\alpha u_0|^2 \exp(2\tau\phi)\,dx$$
$$\leqslant C\tau \int_{\Omega_0} |Pu_0|^2 \exp(2\tau\phi)\,dx$$
$$+ C(\varepsilon)\tau \sum_{|\alpha:m|<1} \int_{\Omega_0 \backslash \Omega_{\varepsilon/2}} |D^\alpha u_0|^2 \exp(2\tau\phi)\,dx.$$

Let $\Phi = \sup \phi$ over Ω_0, then $\phi \leqslant \Phi$ on Ω_0, $\varepsilon < \phi$ on Ω_ε, and $\phi < \varepsilon/2$ on $\Omega_0 \setminus \bar{\Omega}_{\varepsilon/2}$. Replacing ϕ in the left side by its lower bound and in the right side by its upper bound and dividing both sides by $\exp(2\tau\varepsilon)$ we obtain

$$\|u_0\|^2_{m;\,1-1/\mu}(\Omega_\varepsilon) \leqslant C(\varepsilon)\,\tau(\exp(\tau(2\Phi - 2\varepsilon))\,\|Pu_0\|^2_0(\Omega_0) + \exp(-\tau\varepsilon)\,\|u_0\|^2_{m;\,1-1/\mu}(\Omega_0)). \tag{5.5}$$

We claim that

$$\|u_0\|_{m;\,1-1/\mu}(\Omega_\varepsilon) \leqslant C(\varepsilon)\,\|Pu_0\|^\kappa_0(\Omega_0)\,\|u_0\|^{1-\kappa}_{m;\,1-1/\mu}(\Omega_0), \tag{5.6}$$

where $\kappa = \varepsilon/(2\Phi - \varepsilon)$.

If

$$\|u_0\|_{m;\,1-1/\mu} \leqslant \|Pu_0\|_0(\Omega_0)$$

then the estimate (5.6) is obvious. If the opposite inequality is true then we may choose $C(\varepsilon)$ so that

$$\tau = (2/(2\Phi - \varepsilon))\ln(\|u_0\|_{m;\,1-1/\mu}(\Omega_0)/\|Pu_0\|_{m;\,0}) + C(\varepsilon) \geqslant C_1,$$

where C_1 is a constant from Theorem 1.2. With this choice of τ, the first term in the right side of the inequality (5.5) is greater than the second one and inserting this τ we obtain (5.6) as well.

From (5.1), (5.2), (5.3), and (5.6) we have

$$\|u\|_{m;\,1-1/\mu}(\Omega_\varepsilon) \leqslant C\delta + C(\varepsilon)\,\delta^\kappa(\|u\|_{m;\,1-1/\mu}(\Omega_0) + \delta^{1-\kappa}) \tag{5.7}$$

which gives the estimate (1.12).

In the case of the conditions of Theorem 1.1 the operator P is quasi-elliptic, so from the interior estimates of Hörmander [4, Theorem 5.2] applied to Eq. (5.3) where u_0 is extended across Γ_0 as zero using the bounded (5.1) and the representation (5.2) we conclude that

$$\|u\|_{m;\,1}(\Omega_{2\varepsilon}) \leqslant C(\varepsilon)(\delta + \|u\|_{m;\,0}(\Omega_\varepsilon))$$

so from the estimate (5.7) we obtain the estimate (1.11).

The proof is complete.

6. Examples

We give applications to the Cauchy problem with time-like data for some well-known equations.

6.1. *One Space Variable*

Consider the Cauchy problem

$$Pu = f \quad \text{in} \quad \Omega = (0, T) \times (1, 0) \tag{6.1}$$

$$u = g_0, \ldots, (\partial/\partial x_2)^{p-1} u = g_{p-1} \quad \text{on} \quad \Gamma = (0, T) \times \{0\} \tag{6.2}$$

for an linear differential operator P of order p with the "principal part" symbol

$$(1) \quad P_m(\zeta) = -ia\zeta_1^r + \zeta_2^p \qquad \text{or} \qquad (2) \quad P_m(\zeta) = a\zeta_1^r + \zeta_2^p,$$

where $r < p$ and a is a real-valued function in $C^1(\bar{\Omega})$, $a \neq 0$ on $\bar{\Omega}$. Define $m = (r, p)$. We assume that

$$P = P_m + \sum_{|\alpha : m| < 1} a^\alpha D^\alpha, \qquad a^\alpha \in L_\infty(\Omega).$$

Corollary 6.1. *Let Ω_* be any subset of Ω such that $\bar{\Omega}_* \subset \Omega \cup \Gamma$. Then there are constants C, κ, $0 < \kappa < 1$, such that any solution u to the Cauchy problem* (6.1), (6.2), $\|u\|_{m;1}(\Omega) < \infty$, *satisfies the estimate*

$$\begin{gathered} \|u\|_{m;1-1/p}(\Omega_*) \leqslant C(\delta^\kappa \|u\|_{m;1-1/p}^{1-1\backslash\kappa}(\Omega) + \delta) \\ \delta = \|f\|_0(\Omega) + \|g_0\|_{p-1/2}(\Gamma) + \cdots + \|g_{p-1}\|_{1/2}(\Gamma). \end{gathered} \tag{6.3}$$

Proof. We can find a function $x_2 = \omega(x_1)$ such that $\omega \in C^\infty(\mathbb{R})$, $\omega(0) = \omega(T) = 0$, $\Omega_* \subset \{x_2 < \omega(x_1)\}$, and $\omega < 1$. The substitution $y_1 = x_1$, $y_2 = x_2 - \omega(x_1)$ does not change the m-principal part of the operator P, in y-variables $\Omega_* \subset \{y_2 < 0\}$ and Γ is $\partial\Omega_* \cap \{y_2 < 0\}$. Returning to the old notation we may assume that Ω_* belongs to the left half-plane and that Γ is the part of $\partial\Omega$ which is contained is this half-plane.

We are going to show that the function

$$\phi(x) = \exp(-\lambda x_2) - 1$$

satisfies the pseudo-convexity conditions (1.5) ans (1.9) of Theorems 1.1 and 1.2, so Corollary 6.1 follows from Theorem 1.3.

We have $\zeta = (\xi_1, \xi_2 - i\tau\eta_2)$, $\eta_2 = \lambda \exp(-\lambda x_2)$. While checking the pseudo-convexity condition we can assume that P coincides with its m-principal part. The equality $P(\zeta) = 0$ in the case (j) reads

$$(i)^j a\xi_1^r = \zeta_2^p. \tag{6.4}$$

From the assumption $a \neq 0$ on Ω, we conclude that

$$1/C\,|\zeta_2|^{p/r} \leqslant |\xi_1| \leqslant C\,|\zeta_2|^{p/r}. \tag{6.5}$$

The left side in (1.5) is

$$H = \lambda\eta_2\,p^2\,|\zeta_2|^{2(p-1)} + H_1, \tag{6.6}$$

where

$$H_1 = pa_{x_2}\xi_1^r\tau^{-1}\,\mathrm{Im}((i)^j\,(\xi_2 - i\tau\eta_2)^{p-1}).$$

If $j = 1$, from (6.4) we have

$$C\,|\tau| \geqslant |\zeta_2|. \tag{6.7}$$

When proving this inequality we may assume that $|\zeta_2| = 1$, then due to (6.5) we have $1/C \leqslant |\xi_1|$, and since $\tau \neq 0$ for all ξ_1, ζ_2 satisfying (6.4) with $j = 1$ the estimate (6.7) follows by continuity and compactness arguments. From (6.5), (6.6), and (6.7) we derive that

$$|H_1| \leqslant C\,|\zeta_2|^{2p-2}$$

so choosing λ large we obtain from (6.6) that $H > 0$ for all x and all ζ satisfying (6.4).

If $j = 2$, then H_1 is the sum of terms

$$b_l(x)\,\xi_1^r\,\xi_2^{p-2-2l}(\tau\eta_2)^{2l}, \qquad 2 + 2l \leqslant p.$$

From (6.6) it follows that $|H_1| \leqslant C\eta_2\,|\zeta_2|^{2p}$, so again choosing λ large we conclude that $H > 0$. Observing that the left side in the inequality (1.9) is the limit of H as τ goes to 0 we obtain both pseudo-convexity conditions of Theorems 1.1 and 1.2. Now Corollary 6.1 follows from Theorem 1.3.

The following differential operators P satisfy the conditions of Corollary 6.1 (we let $x_1 = t$, $x_2 = x$)

$$au_t + u_{xx} + bu_x + cu \qquad \text{(parabolic operators)}$$

$$iau_t + u_{xx} + bu_x + cu \qquad \text{(Schrödinger operators)}$$

$$au_t + u_{xxx} + bu_{xx} + cu_x + du \qquad \text{(Korteweg–de Vries type operators)}$$

$$au_{tt} + u_{xxxx} + bu_{tx} + cu_{xxx} + \cdots + hu,$$

where $a \in C^1(\bar{\Omega})$, $0 < a$ on $\bar{\Omega}$, b, $c, ..., h$ are bounded and measurable functions on Ω.

In what follows we let $\Omega=(0, T)\times G$ where G is a bounded domain in $\mathbb{R}^n$ and let $\Gamma=(0, T)\times\gamma$ where γ is a C^2-smooth open part of ∂G. For a second order operator P we consider the Cauchy problem

$$Pu=f \qquad \text{on } \Omega \tag{6.8}$$

$$u=g_0, \qquad \partial u/\partial\nu=g_1 \text{ on } \Gamma. \tag{6.9}$$

Let $m=(1, 2, ..., 2)$.

6.2. *Second Order Parabolic Operators*

$$P(\zeta)=i\zeta_1+\sum a^{jk}(x)\,\zeta_j\zeta_k+\sum a^j(x)\,\zeta_j+a(x),$$

where the coefficients a^{jk} are real-valued and belong to $C^1(\bar{\Omega})$, a^j, $a\in L_\infty(\Omega)$, and the sums are over $j, k=2, ..., n$. We assume that the following parabolicity condition is satisfied

$$\varepsilon(\xi_2^2+\cdots+\xi_n^2)\leqslant\sum a^{jk}(x)\,\xi_j\xi_k \qquad \text{for all} \quad x\in\Omega,\ \xi\in\mathbb{R}^{n-1}. \tag{6.10}$$

COROLLARY 6.2. *Let Ω_* be any open subset of Ω such that $\bar{\Omega}_*\subset\Omega\cup\Gamma$. Then there are constants C, κ, $0<\kappa<1$, such that any solution u to the Cauchy problem* (6.8), (6.9), $\|u\|_{m,1}<\infty$, *satisfies the estimate*

$$\|u\|_{m;1}(\Omega_*)\leqslant C(\delta^\kappa\,\|u\|_0^{1-\kappa}(\Omega)+\delta), \tag{6.11}$$

where

$$\delta=\|f\|_0(\Omega)+\|g_0\|_{3/2}(\Gamma)+\|g_1\|_{1/2}(\Gamma).$$

Proof. Observe that Ω_* can be covered by a finite number of the images of the half-cylinder $\Omega_C=\{0<y_1<T,\quad y_2^2+\cdots+y_n^2<1,\ 0<y_n\}$ under diffeomorphisms $x_1=y_1$, $x_j=x_j(y_2, ..., y_n)$, $2\leqslant j$, of class $C^\infty(\Omega_C)$ so that the images of $\partial\Omega_C\cap\{y_n=0\}$ belong to Γ. Such diffeomorphisms do not change the form of the operator P, so it is sufficient to consider $\Omega=\Omega_C$ only. Moreover, by using another substitution $x_1=y_1, ..., x_{n-1}=y_{n-1}$, $x_n=y_n+h(y_1, ..., y_{n-1})$ similar to those described in the proof of Corollary 6.1 we can always reduce Ω to a subdomain of the half-space $\{x_n<0\}$ so that $\Gamma=\partial\Omega\cap\{x_n<0\}$. So we consider such a domain Ω.

We will show that the function $\phi(x)=\exp(-\lambda x_n)-1$ again satisfies the pseudo-convexity condition of Theorem 1.1.

We have $q=2$, $\nabla_q \phi=(0, \ldots, 0, -\eta_n)$, $\eta_n=\lambda \exp(-\lambda x_n)$. Further we let $P=P_m$. The equality $P(x; \zeta)=0$, $\zeta=\xi+i\tau\nabla_q\phi$ is equivalent to the relations

$$\sum_{1\leqslant j,k\leqslant n} a^{jk}(x)\,\xi_j\xi_k=\tau^2 a^{nn}(x)\,\eta_n^2$$
$$\xi_1=2\sum_{2\leqslant j\leqslant n}\tau a^{jn}(x)\,\xi_j\eta_n. \tag{6.12}$$

Due to the condition (6.10) we have then $|'\xi|\leqslant C\eta_n$ where $'\xi=(0, \xi_2, \ldots, \xi_n)$. The left side H in (1.5) is the sum of the term

$$4\lambda\eta_n\left(\left(\sum_{2\leqslant j\leqslant n} a^{jn}\xi_j\right)^2+(a^{nn})^2\,\tau^2\eta_n^2\right)\geqslant C^{-1}\tau^2\lambda\eta_n^3$$

and of terms

$$\tau^{-1}b(x)\,\mathrm{Im}\,\zeta_j\zeta_k\bar{\zeta}_l,\qquad 2\leqslant j,\ k,\ l\leqslant n,$$

where $b(x)$ is a $L_\infty(\Omega)$-function, these terms are either $b(x)\,\xi_j\xi_k\eta_n$ or $b(x)\,\tau^2\eta_n^3$, so they are bounded by $C\tau^2\eta_n^3$. Finally,

$$H\geqslant C^{-1}\tau^2\lambda\eta_n^3-C\tau^2\eta_n^3.$$

By choosing λ large we guarantee that $H>0$.

Now, as above, Corollary 6.2 follows from Theorem 1.3.

6.3. *The Schrödinger Operator*

$$P(\zeta)=a\zeta_1+\sum_{2\leqslant j\leqslant n}\zeta_j^2+\sum_{2\leqslant j\leqslant n} a^j(x)\,\zeta_j+a(x),$$

where $a\in C^1(\bar{\Omega})$, $0<a$ on $\bar{\Omega}$, a^j, $a\in L_\infty(\Omega)$. Here we assume that $G\subset\{x_n<0\}$ and $\gamma=\partial G\cap\{x_n<0\}$. Actually, in the most interesting case $G=G_*\cap\{x_n<0\}$ where G_* is a domain with C^2-smooth boundary, so ∂G consists of $\gamma\subset\{x_n<0\}$ and of a part which belongs to $\{x_n=0\}$.

COROLLARY 6.3. *Suppose that the coefficient a satisfies the conditions*

$$-2a<\sum_{2\leqslant j\leqslant n-1}\partial a/\partial x_j x_j,\qquad \partial a/\partial x_n\leqslant 0 \text{ on } \Omega. \tag{6.13}$$

Then the claim of Corollary 6.2 is valid with the estimate

$$\|u\|_{m;1/2}(\Omega_*)\leqslant C(\delta^\kappa\,\|u\|_{m;1/2}^{1-\kappa}(\Omega)+\delta) \tag{6.14}$$

instead of the estimate (6.11).

Proof. As in the proof of Corollary 6.1, for any Ω_* we can find a substitution

$$x_1 = y_1, \ldots, x_{n-1} = y_{n-1}, \qquad x_n = y_n + \omega(y_1) \tag{6.15}$$

such that $\Omega_* \subset \{y_n < -2\varepsilon\}$ while Γ (in y-variables) contains $\partial\Omega \cap \{y_n < 0\}$. Since the m-principal part of P as well as the condition (6.13) does not change under this substitution we may assume that $\Gamma = \partial\Omega \cap \{x_n < 0\}$.

We will show that the function

$$\phi(x) = \psi(x) - 1,$$
$$\psi(x) = \exp(\lambda/2((x_1 - T/2)^2 + x_2^2 + \cdots + x_{n-1}^2 + (x_n - \sigma)^2 - 1 - \sigma^2))$$

for large λ, σ satisfies the pseudo-convexity conditions of Theorems 1.1, 1.2, so Corollary 6.3 will follow from Theorem 1.3.

We have

$$\nabla_q \phi = \lambda\psi x(\sigma), \qquad x(\sigma) = (x_2, \ldots, x_{n-1}, x_n - \sigma)$$
$$\partial^2\phi/\partial x_j\, \partial x_k = \lambda\delta_{jk}\psi + \lambda^2 x_j(\sigma)\, x_k(\sigma)\psi.$$

The equality $P(\zeta) = 0$ for $\zeta = \xi + i\tau\lambda\psi x(\sigma)$ is equivalent to the relations

$$a\xi_1 + \sum \xi_j^2 - \tau^2\lambda^2\, |x(\sigma)|^2\, \psi^2 = 0 \tag{6.16}$$

$$\tau \sum \xi_j x_j(\sigma) = 0. \tag{6.17}$$

Everywhere in this proof the sums are over $j = 2, \ldots, n$. The left side H in (1.5) is

$$\begin{aligned} H &= 4\lambda\left(\sum \xi_j^2 + \tau^2\lambda^2\, |x(\sigma)|^2\, \psi^2\right) + 4\lambda^2 \left|\sum x_j(\sigma)\, \zeta_j\right|^2 - 2\xi_1\lambda \sum \partial a/\partial x_j x_j(\sigma) \\ &= 2\lambda\left(2 + a^{-1} \sum \partial a/\partial x_j x_j(\sigma)\right) \sum \xi_j^2 \\ &\quad + \left(2 + 2\lambda\, |x(\sigma)|^2 - a^{-1} \sum \partial a/\partial x_j x_j(\sigma)\right) 2\tau^2\lambda^3\, |x(\sigma)|^2\, \psi^2 \end{aligned} \tag{6.18}$$

if we express ξ_1 from (6.16), assume that $\tau \neq 0$, and use that due to (6.17) we have $x_2(\sigma)\, \zeta_2 + \cdots + x_n(\sigma)\, \zeta_n = i\tau\lambda\psi\, |x(\sigma)|^2$.

The sign of

$$2 + a^{-1} \sum_{2 \leq j \leq n} \partial a/\partial x_j x_j(\sigma) = 2 + a^{-1} \sum \partial a/\partial x_j x_j + a^{-1}\, \partial a/\partial x_n (x_n - \sigma)$$

can be made positive if we use the condition (6.13) and choose σ large enough. Henceforth we fix σ. Since σ is fixed we have $|x(\sigma)| > 1/C$ on Ω, so the last term in the expression (6.18) for H can be made positive by choosing λ large.

We have $\lim H > 0$ as τ goes to 0 as well.

As observed above this implies that the function ϕ is pseudo-convex, so we may apply Theorem 1.3. Taking into account that $\phi > 0$ if $x_n < h$ where $h = -\sigma((1+\sigma^{-2})^{1/2} - 1) \geq -1/2\sigma$ (actually, the point $(0, ..., 0, h)$ is the point of the intersection of the sphere $\{\phi = 0\}$ with the x_n-axis) and choosing σ large enough we achieve that $\Omega_* \subset \Omega_0 \cup \Gamma$, so Corollary 6.3 follows from Theorem 1.3.

Some results of the book of Zuily [16, Remark 2.7, p. 127] suggest that Corollary 6.3 is somehow sharp: if $\partial a/\partial x_n > 0$ or if Ω is not the convex hull of Γ then non-uniqueness occurs.

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References

1. T. Carleman, Sur un problème d'unicité pour les systèmes d'équations aux derivées partielles à deux variables independantes, *Ark. Mat. Astr. Fys. B* **26** (1939), 1–9.
2. B. Dehman, Unicité du problème de Cauchy pour une classes d'opérateurs quasi-homogeneous, *J. Math. Kyoto Univ.* **24** (1984), 453–471.
3. L. Hörmander, "Linear Partial Differential Operators," Springer, New York, 1963.
4. L. Hörmander, On interior regularity of the solutions of partial differential equations, *Comm. Pure Appl. Math.* **11** (1958), 197–218.
5. V. Isakov, On the uniqueness of the solution of the Cauchy Problem, *Soviet Math. Dokl.* **22** (1980), 639–642.
6. V. Isakov, A non-hyperbolic Cauchy problem for $\square_b \square_c$ and its applications to elasticity theory, *Comm. Pure Appl. Math.* **41** (1986), 747–769.
7. V. Isakov, Uniqueness of the continuation across a time-like hyperplane and related inverse problems for hyperbolic equations, *Comm. Partial Differential Equations* **14** (1989), 465–478.
8. V. Isakov, "Inverse Source Problems," Math. Surveys Monographs, Amer. Math. Soc., Providence, RI, 1990.
9. F. John, Continuous dependence on data for solutions of partial differential equations with a prescribed bound, *Comm. Pure Appl. Math.* **13** (1960), 551–585.
10. H. Khalgui-Ounaies, Unicité de problème de Cauchy pour les opérateurs quasi-homogenes, *Ann. Scuola Norm. Sup. Pisa* **15** (1988), 567–583.
11. P. Knabner and S. Vessella, Stabilization of ill-posed Cauchy problems for parabolic equations, *Ann. Mat. Pura Appl.* **149** (1987), 393–409.

12. H. KUMANO-GO, Pseudo-differential operators and the uniqueness in the Cauchy problem, *Comm. Pure Appl. Math.* **22** (1969), 73–129.
13. J.-L. LIONS AND E. MAGENES, "Non-Homogeneous Boundary Value Problems, I," Springer, New York, 1972.
14. L. NIRENBERG, Uniqueness in Cauchy problems for differential operators with constant leading coefficients, *Comm. Pure Appl. Math.* **10** (1957), 89–105.
15. J.-C. SAUT AND B. SCHEURER, Unique continuation for some evolution equations, *J. Differential Equations* **66** (1987), 118–139.
16. C. ZUILY, "Uniqueness and Non-Uniqueness in the Cauchy Problem," Birkhäuser, Boston, 1983.